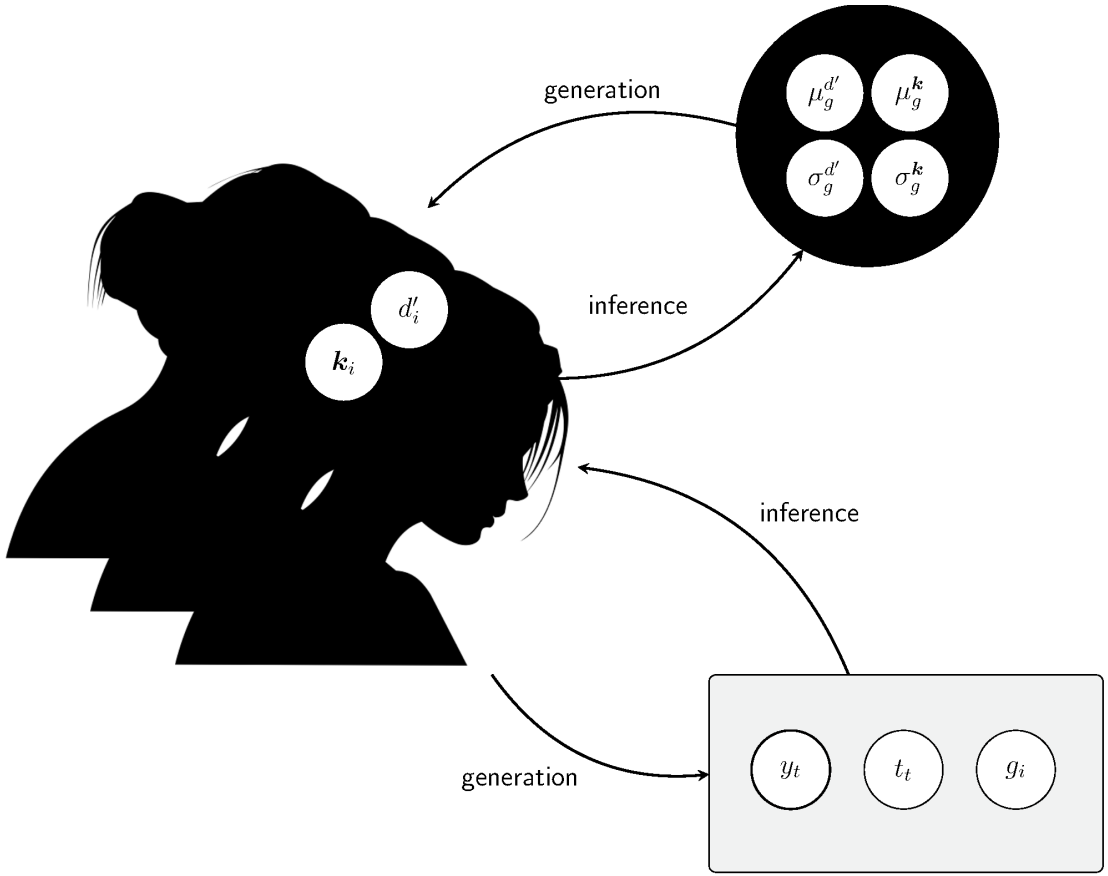
Hierarchical Multiple Choice Signal Detection Model

Michael D. Lee

Hierarchical Multiple Choice Signal Detection Model Structure

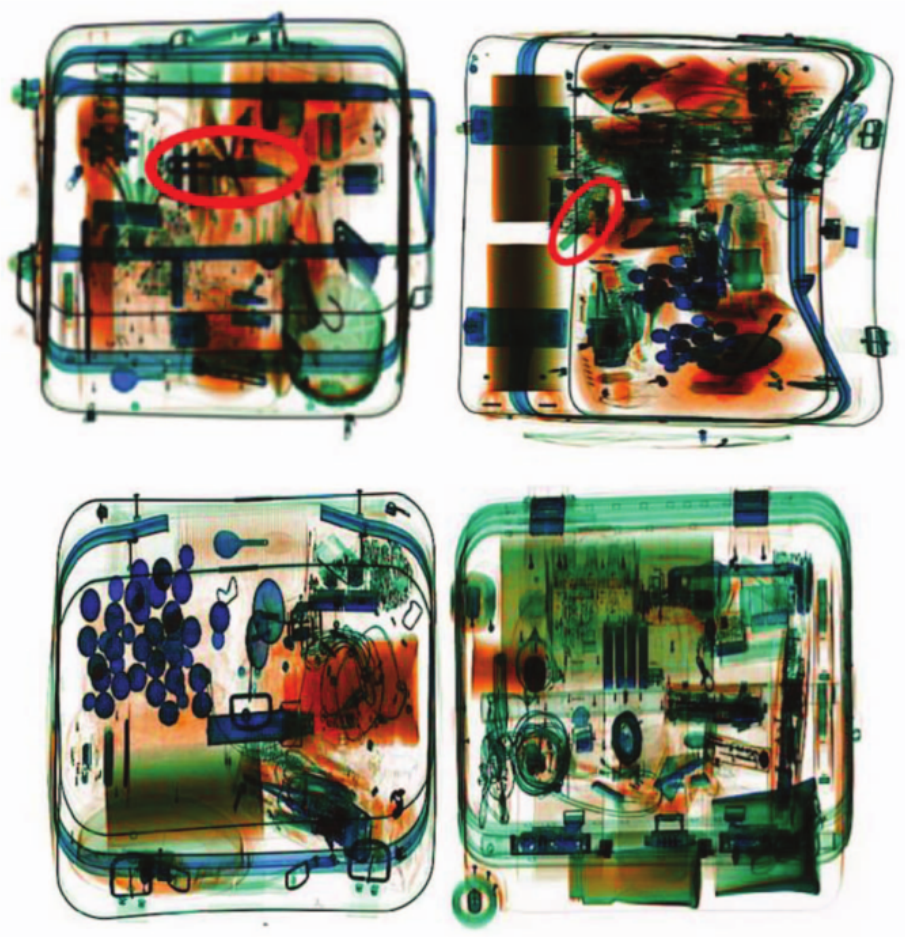
- The hierarchical multiple choice signal detection model has the structure
  - observed binary decisions y_t on trial t with the truth t_t indicating signal or noise
  - discriminability d' and a vector of criteria $\boldsymbol{k} = (k_1, k_2, \ldots)$ parameters



Baggage Screening

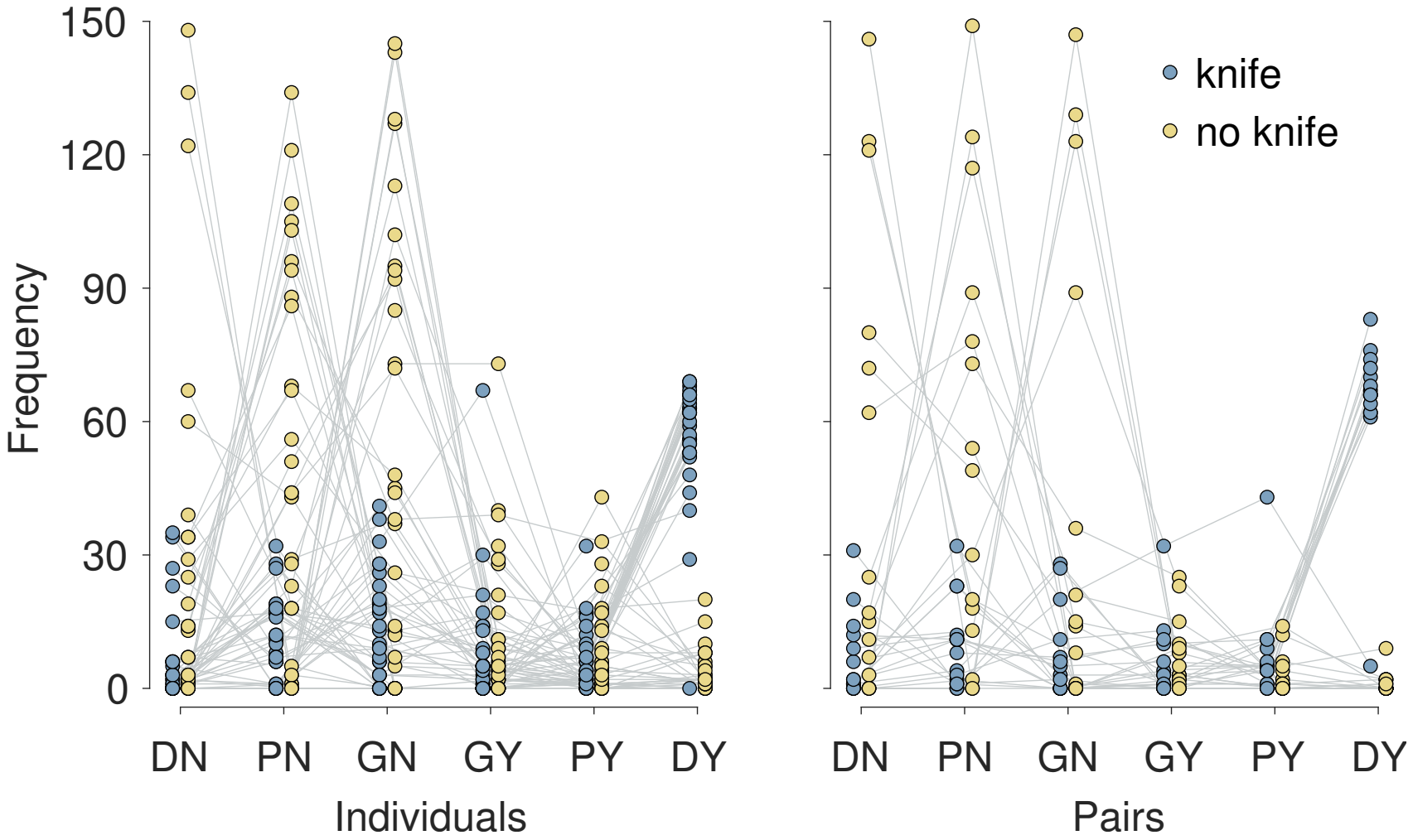
Baggage Screening Experiment

- Enright and McCarley (2019) studied people's decision making in a mock baggage screening task
  - task is to detect a knife in an x-ray image of baggage
  - participants did the task both individually and in pairs



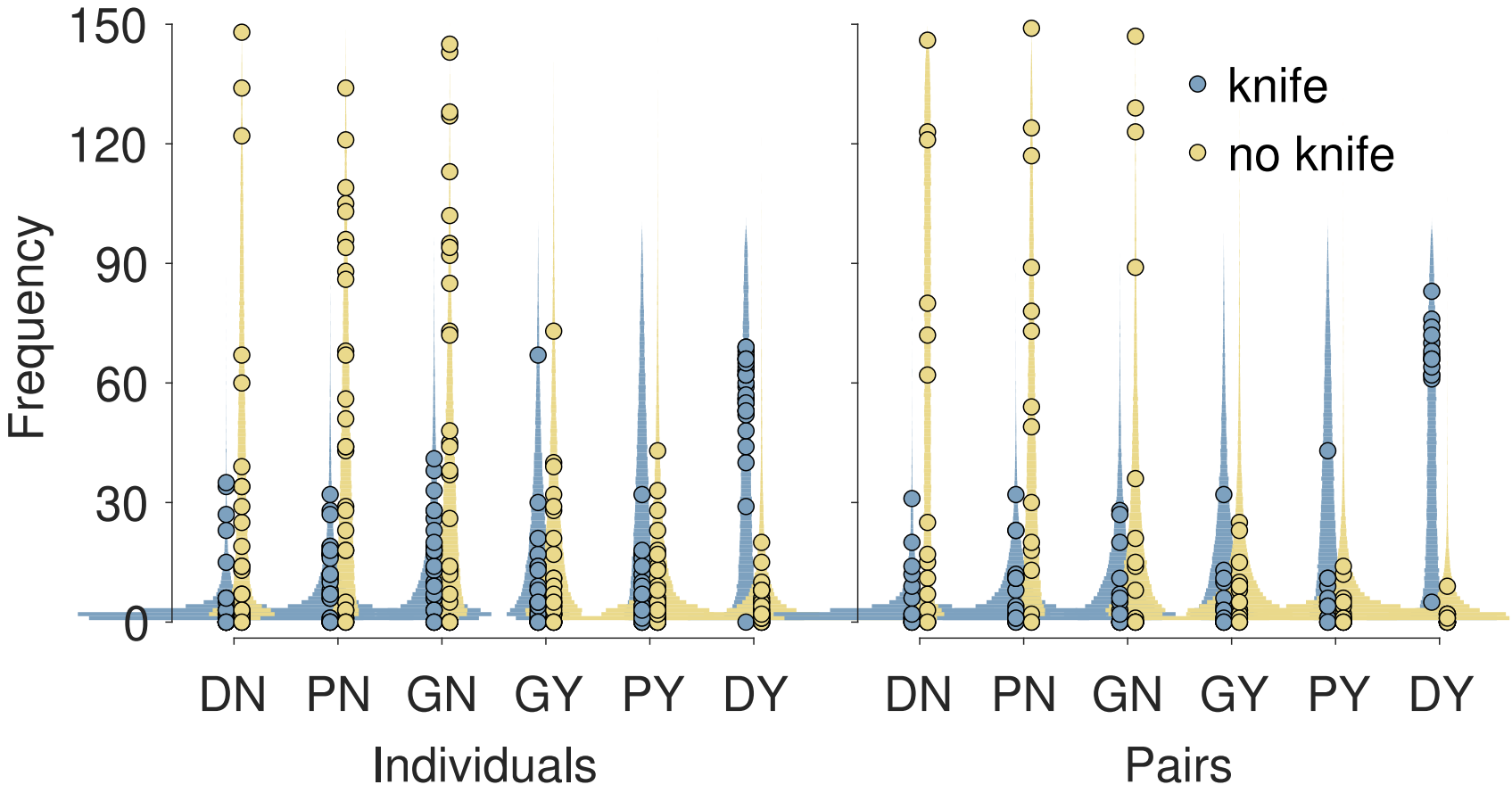
Baggage Screening Data

- There were 250 trials with 100 containing a knife
  - 6-point response scale with labels "definitely no," "probably no," "guess no," "guess yes," "probably yes," and "definitely yes."



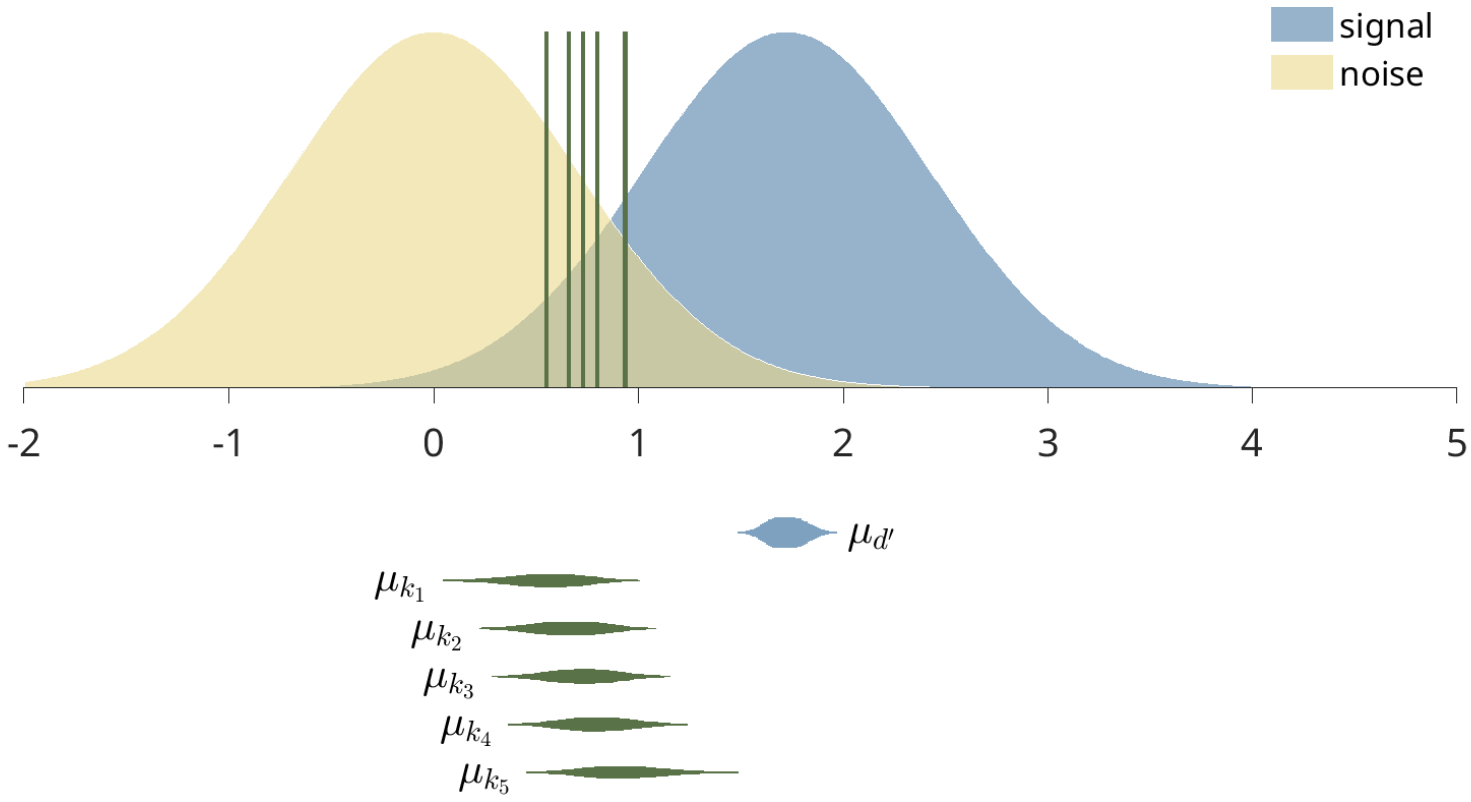
Descriptive Adequacy

- Posterior predictive distribution for the representative individuals in each group captures important regularities of the behavioral data
  - expect "no" responses to images with knives to be rare
  - allow for a wide range of frequencies for "no" responses with different levels of confidence when images do not have knives.



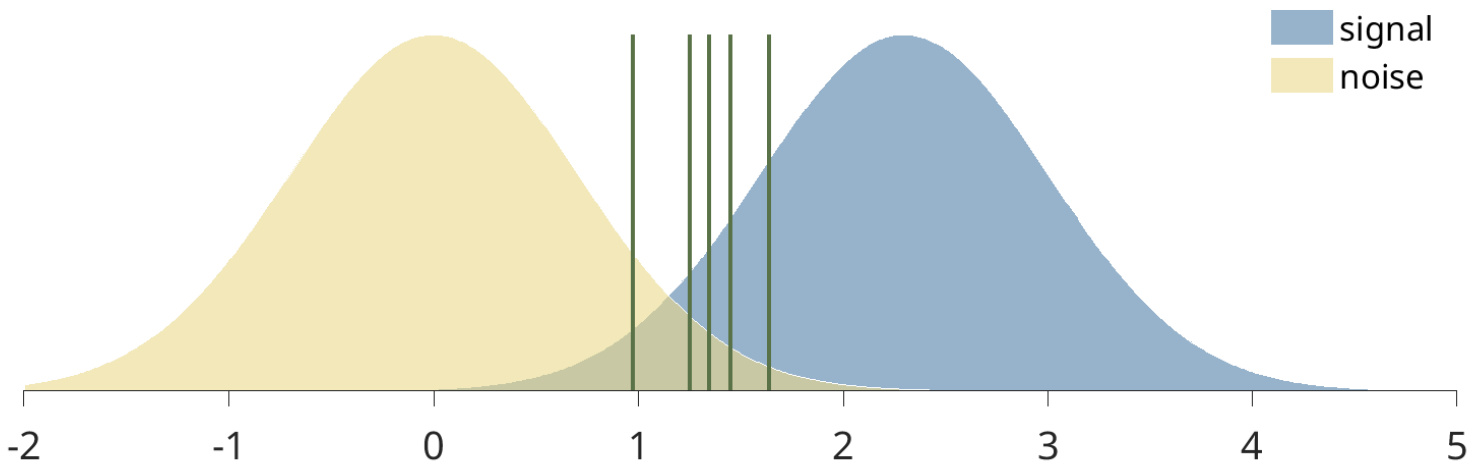
Group Inference for Individuals Completing Task

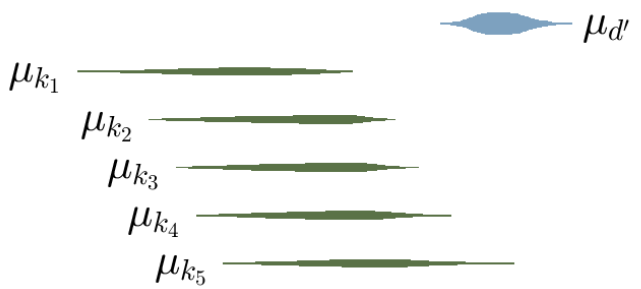
- Mean discriminability for the individuals is $\mu_{d'} = 1.73$ with 95% credible interval $(1.59, 1.86)$
  - lots of uncertainty about criteria placement



Group Inference for Pairs Completing Task

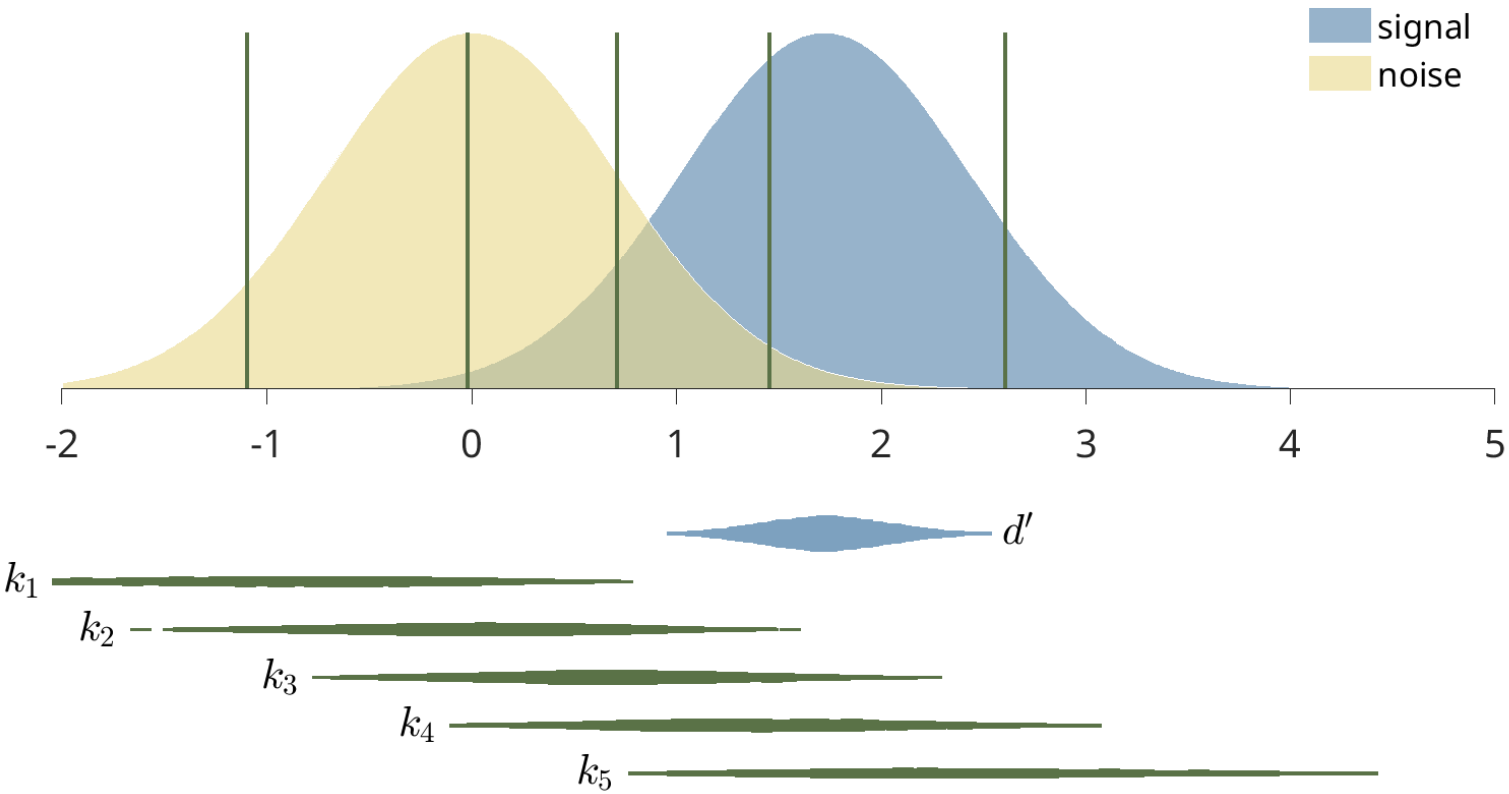
- Mean discriminability for the pairs is $\mu_{d'} = 2.31$ with 95% credible interval $(2.12, 2.52)$
  - better ability to detect knives
  - rightward shift of (highly uncertain) criteria suggests that more evidence is needed to respond "knife"





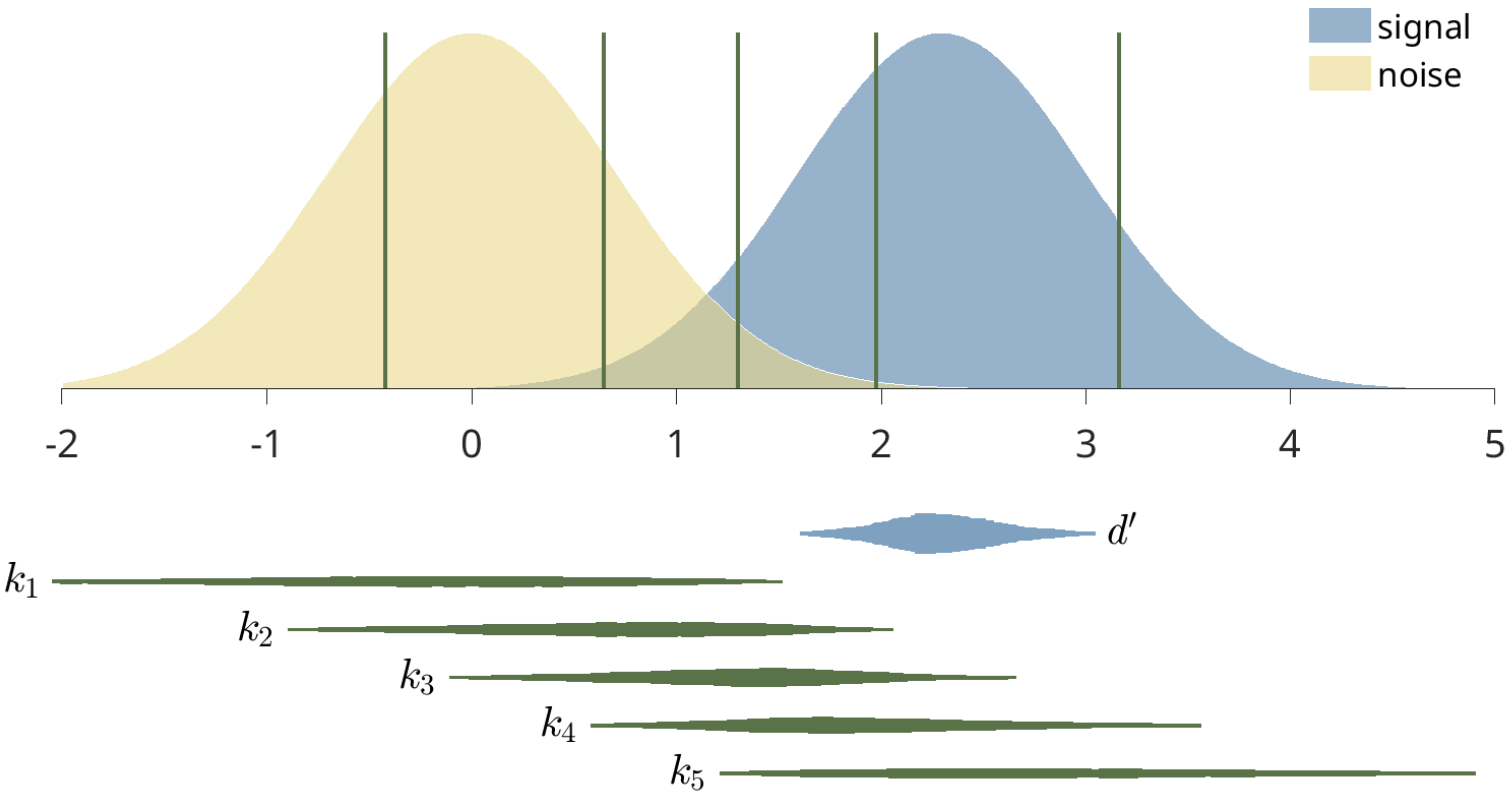
Representative Individual Inference for Individuals Completing Task

- There is much more uncertainty for the representatitive individual of individuals completing the task
  - has to accommodate individual differences
  - there is very little information in the inferences about the criteria



Representative Individual Inference for Pairs Completing Task

- Some evidence at the representative individual level that pairs have better discriminability
  - but more uncertainty than at the group level
  - criteria could be regarded as nuisance parameters needed so that inferences about discriminability are based on reasonable assumptions



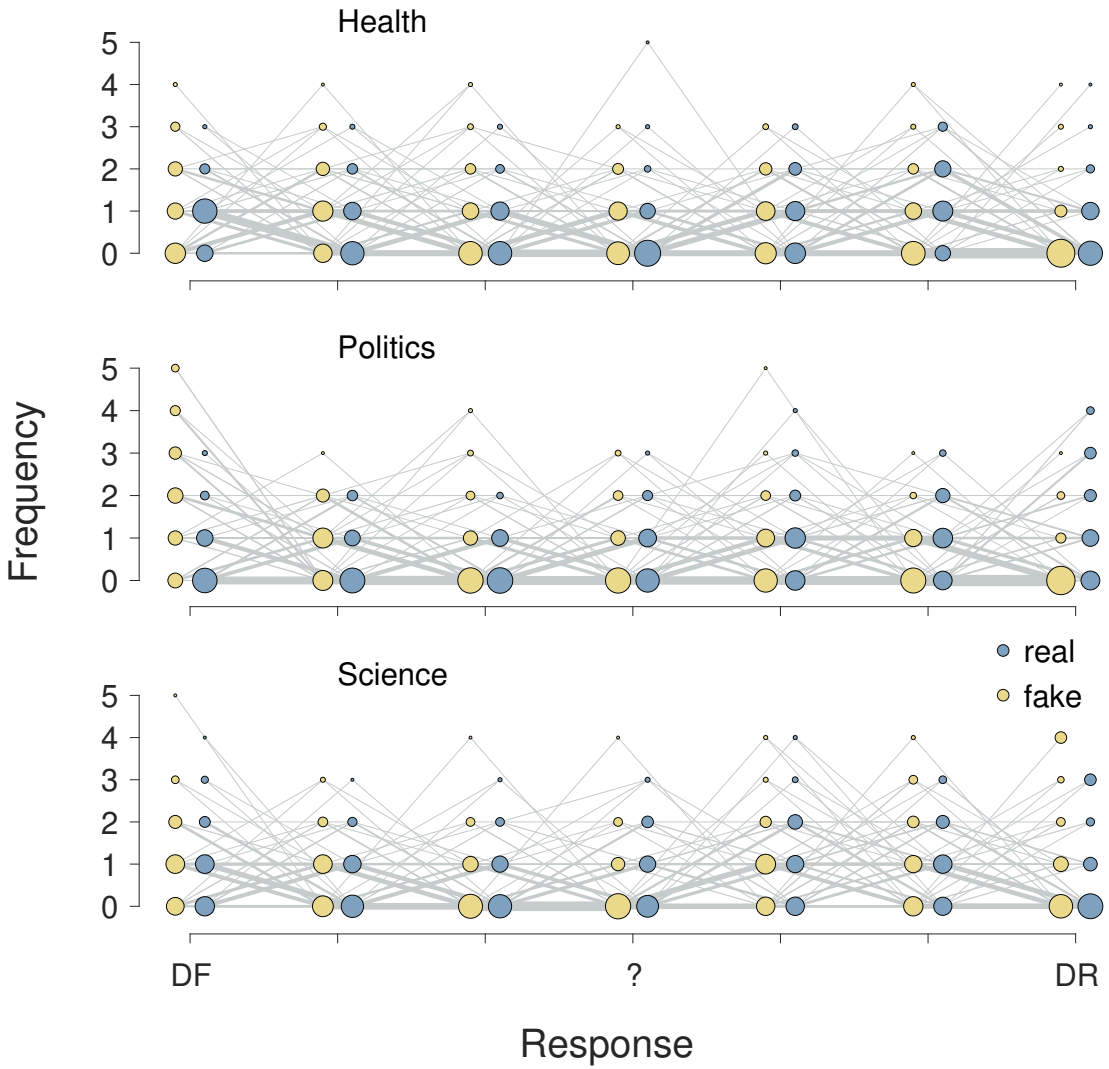
Fake News

Fake News Experiment

- Luo, Hancock, and Markowitz (2022, Experiment 1) studied people's ability to distinguish between real and fake news headlines
  - headlines weere sourced from social media
  - 337 participants were randomly assigned to health, politics, or science content area
  - each participant was shown ten headlines, five of which were real and five or which were fake
  - responded on a seven-point scaling ranging from 1 = "definitely fake" to 7 = "definitely real"

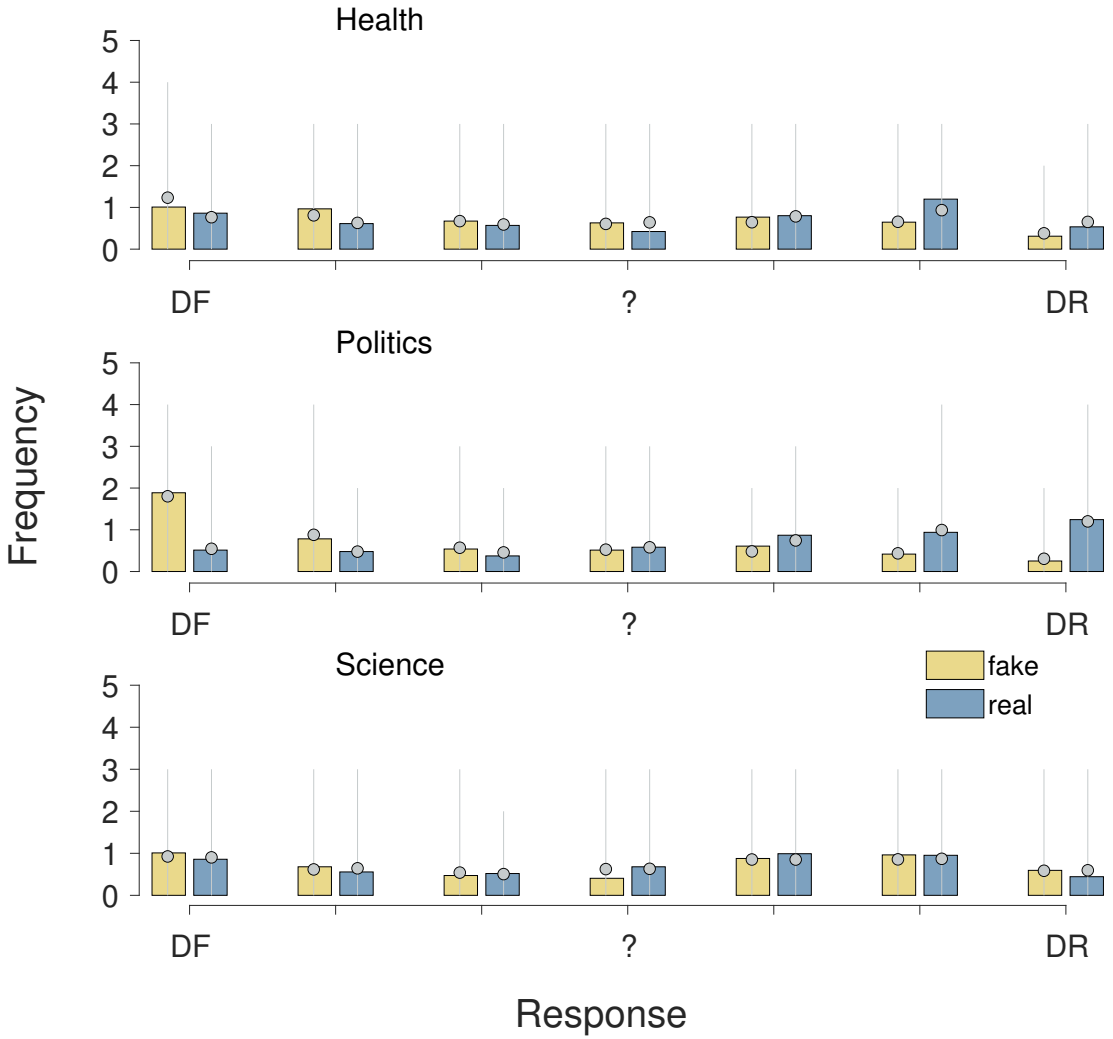
Fake News Data

- An attempt at summarizing the responses for all content areas and all participants is shown



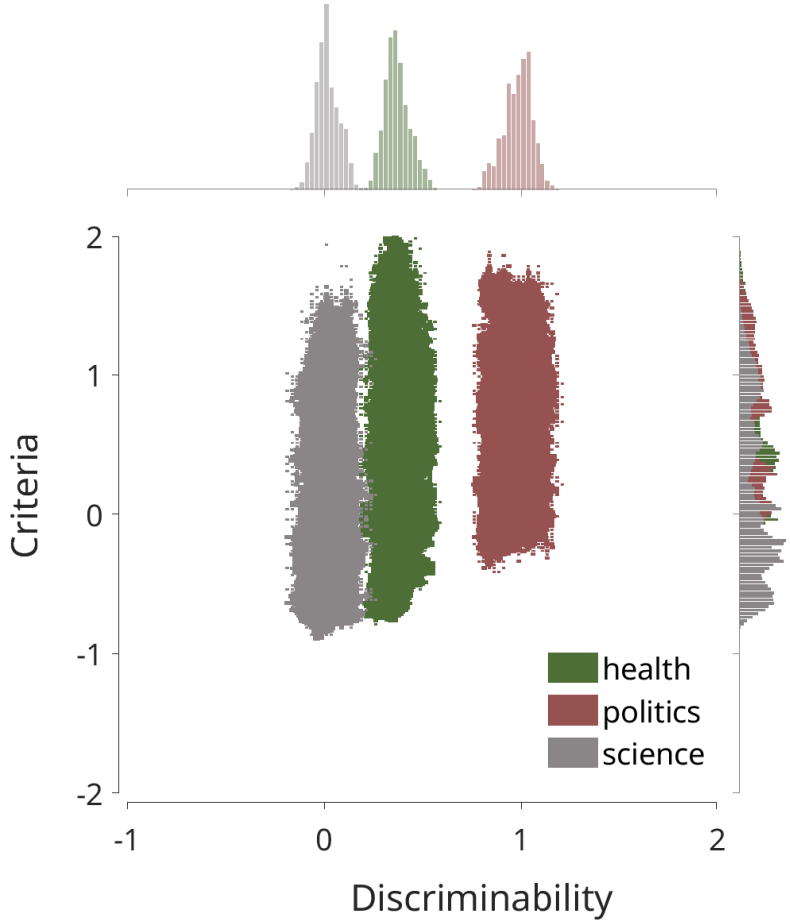
Descriptive Adequacy

- The posterior predictive distribution is overlayed on a simpler summary of the data that aggregates over individual participants



Group Inferences

- The joint group posterior distribution has lots of information
  - the answer to the motivating scientific question is clear in terms of content-area differences in detecting fake news
  - a good example of using hierarchical models when many participants each provide few data



References i

Enright, A., & McCarley, J. S. (2019). Collaborative search in a mock baggage screening task. *Journal of experimental psychology: Applied*, *25*(4), 716. doi: 10.31234/osf.io/8975x

Luo, M., Hancock, J. T., & Markowitz, D. M. (2022). Credibility perceptions and detection accuracy of fake news headlines on social media: Effects of truth-bias and endorsement cues. *Communication Research*, *49*(2), 171–195. doi: 10.1177/0093650220921321